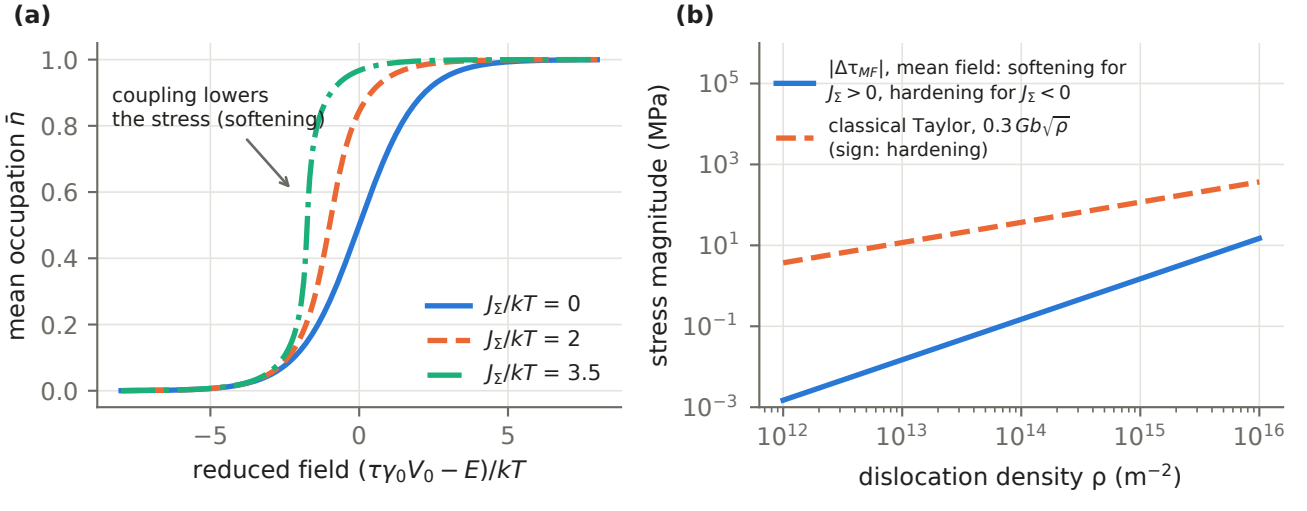
(a)
mean occupation $\bar{n}$
1.0
0.8
0.6
0.4
0.2
0.0
coupling lowers
the stress (softening)
$J_\Sigma/kT = 0$
$J_\Sigma/kT = 2$
$J_\Sigma/kT = 3.5$
−5
0
5
reduced field $(\tau\gamma_0 V_0 - E)/kT$
(b)
stress magnitude (MPa)
10^5
10^3
10^1
10^{-1}
10^{-3}
$|\Delta\tau_{MF}|$, mean field: softening for
$J_\Sigma > 0$, hardening for $J_\Sigma < 0$
classical Taylor, $0.3\,Gb\sqrt{\rho}$
(sign: hardening)
10^{12}
10^{13}
10^{14}
10^{15}
10^{16}
dislocation density ρ (m^{-2})